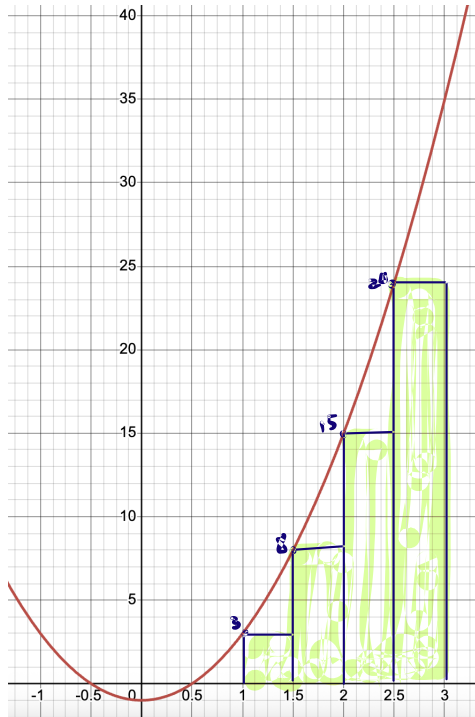




1. Consider the function $f(x) = 4x^2 - 1$ on the interval $[1, 3]$. The graph of the function is given below.

- 1). Sketch the appropriate rectangles on the graph to illustrate the left-endpoint Riemann sum L_4 for $f(x)$ on the interval $[1, 3]$ with $n = 4$.
- 2). Calculate the left-endpoint Riemann sum L_4 for $f(x)$ on the interval $[1, 3]$ with $n = 4$.



$$\Delta x = \frac{3-1}{4} = \frac{1}{2}$$

Subintervals: $[1, \frac{3}{2}]$, $[\frac{3}{2}, 2]$, $[2, \frac{5}{2}]$, $[\frac{5}{2}, 3]$

Left-endpoints: 1, $\frac{3}{2}$, 2, $\frac{5}{2}$

Height of rectangles:	$f(1)$	$f(\frac{3}{2})$	$f(2)$	$f(\frac{5}{2})$
	$= 4 \cdot 1^2 - 1$	$= 4 \cdot (\frac{3}{2})^2 - 1$	$= 4 \cdot 2^2 - 1$	$= 4 \cdot (\frac{5}{2})^2 - 1$
	$= 3$	$= 4 \cdot \frac{9}{4} - 1$	$= 16 - 1$	$= 4 \cdot (\frac{25}{4}) - 1$
		$= 8$	$= 15$	$= 24$

(or: one may use the graph to figure out that

$f(1)=3$, $f(\frac{3}{2})=8$, $f(2)=15$, $f(\frac{5}{2})=24$, as shown in the graph)

$$L_4 = f(1)\Delta x + f(\frac{3}{2})\Delta x + f(2)\Delta x + f(\frac{5}{2})\Delta x$$

$$= 3 \cdot \frac{1}{2} + 8 \cdot \frac{1}{2} + 15 \cdot \frac{1}{2} + 24 \cdot \frac{1}{2}$$

$$= \frac{3}{2} + 4 + \frac{15}{2} + 12$$

$$= 25$$